

MODULE 6: ELECTRONIC MAIL SECURITY

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Electronic Mail

Electronic Mail (E-mail) is one of the most widely used communication methods on the Internet. Since e-mails often contain sensitive information such as personal data, financial records, and business documents, protecting them from unauthorized access and modification is essential. Electronic Mail Security provides mechanisms to ensure that messages remain confidential, authentic, and unaltered during transmission.

Basics of Mail Security

Mail Security refers to the techniques and protocols used to protect e-mail messages against threats such as interception, modification, impersonation, and unauthorized access.

Security Requirements of E-Mail

- 1. Confidentiality:** Confidentiality ensures that only the intended recipient can read the message contents. If an attacker intercepts the e-mail, the information remains unreadable due to encryption.
- 2. Authentication:** Authentication verifies the identity of the sender and assures the receiver that the message actually originated from the claimed sender.
- 3. Integrity:** Integrity guarantees that the message has not been altered during transmission. Any modification made by an attacker can be detected using hash functions and digital signatures.
- 4. Non-Repudiation:** Non-repudiation prevents the sender from denying that the message was sent. Digital signatures provide proof of origin.
- 5. Availability:** Availability ensures that e-mail services remain accessible to legitimate users whenever required.

Threats to E-Mail Security

- 1. Eavesdropping:** An attacker secretly intercepts and reads e-mail messages while they are being transmitted.
- 2. Message Modification:** The contents of an e-mail are changed by an attacker before reaching the recipient.
- 3. Spoofing:** An attacker impersonates another user by forging the sender's e-mail address.
- 4. Replay Attack:** Previously captured e-mails are retransmitted to create confusion or gain unauthorized access.
- 5. Malware Attack:** Viruses, worms, and ransomware are distributed through malicious e-mail attachments.
- 6. Phishing:** Fraudulent e-mails trick users into revealing passwords, banking information, or other confidential data.

Security Mechanisms Used in E-Mail

- Symmetric Key Encryption
- Public Key Encryption
- Digital Signatures
- Hash Functions
- Digital Certificates
- Public Key Infrastructure (PKI)

Pretty Good Privacy (PGP)

Pretty Good Privacy (PGP) is a popular e-mail security system that provides secure communication by combining symmetric encryption, public-key encryption, compression, and digital signatures. PGP is known as a **hybrid cryptosystem** because it uses both symmetric and asymmetric encryption techniques. **Objectives of PGP**

1. Protect message confidentiality.
2. Verify sender authenticity.
3. Ensure message integrity.
4. Provide non-repudiation.
5. Improve transmission efficiency through compression.

Services Provided by PGP

1. **Authentication:** PGP uses digital signatures to verify the sender's identity and ensure the message originates from the claimed sender.
2. **Confidentiality:** The message is encrypted using a session key, ensuring that only authorized recipients can read it.
3. **Compression:** Before encryption, the message is compressed to reduce its size and improve transmission efficiency.
4. **E-Mail Compatibility:** PGP converts binary encrypted data into ASCII format using Radix-64 encoding, making it suitable for e-mail transmission.
5. **Segmentation:** Large messages can be divided into smaller segments for easier transmission.

PGP Encryption Process

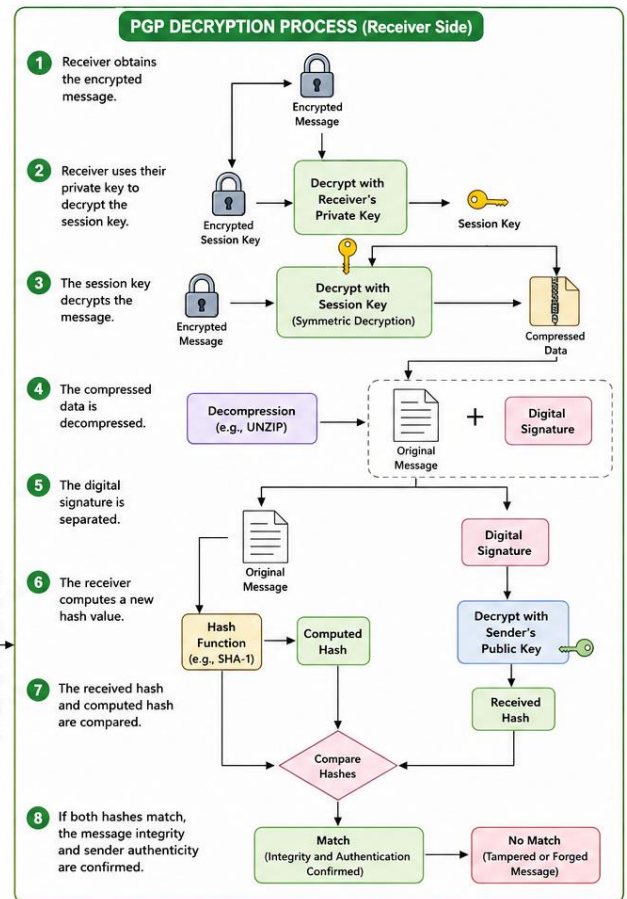
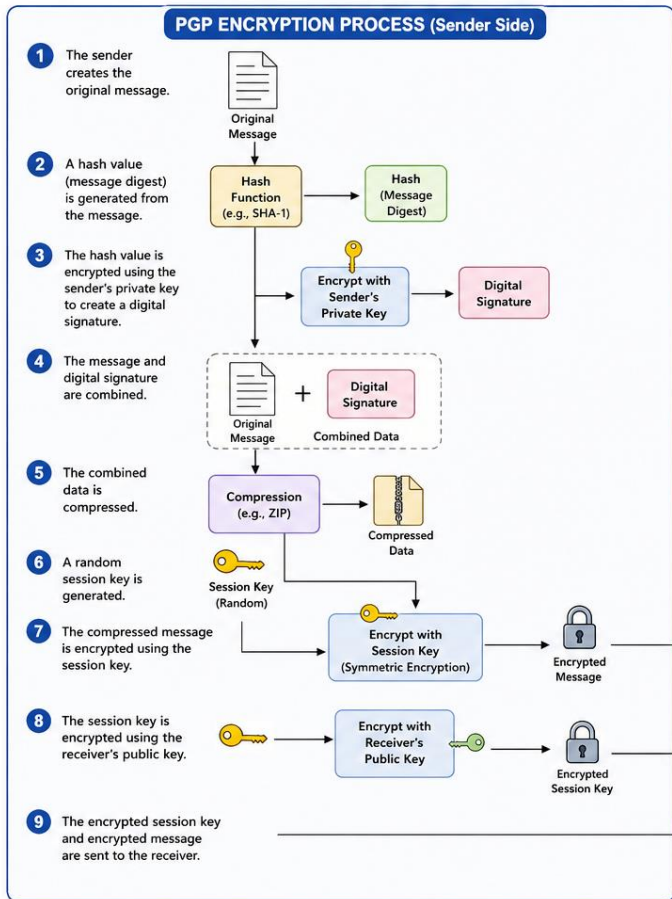
1. The sender creates the original message.
2. A hash value (message digest) is generated from the message.
3. The hash value is encrypted using the sender's private key to create a digital signature.
4. The message and digital signature are combined.
5. The combined data is compressed.
6. A random session key is generated.
7. The compressed message is encrypted using the session key.
8. The session key is encrypted using the receiver's public key.

9. The encrypted session key and encrypted message are sent to the receiver.

PGP Decryption Process

1. Receiver obtains the encrypted message.
2. Receiver uses their private key to decrypt the session key.
3. The session key decrypts the message.
4. The compressed data is decompressed.
5. The digital signature is separated.
6. The receiver computes a new hash value.
7. The received hash and computed hash are compared.
8. If both hashes match, the message integrity and sender authenticity are confirmed.

PGP PROCESS – ENCRYPTION AND DECRYPTION



Advantages of PGP

- Strong security.
- Provides confidentiality and authentication.
- Efficient due to hybrid encryption.
- Supports digital signatures.
- Widely used for secure communication.

Disadvantages of PGP

- Complex key management.
- Difficult for non-technical users.
- Trust establishment can be challenging.

Web of Trust

Unlike certificate-based systems, PGP uses a **Web of Trust** model. In this model:

- Users sign each other's public keys.
- Trust is established through recommendations.
- No central authority is required.

This decentralized approach makes PGP flexible but sometimes difficult to manage.

S/MIME (Secure/Multipurpose Internet Mail Extensions)

S/MIME (Secure Multipurpose Internet Mail Extensions) is a standard for secure electronic mail that uses public key cryptography and digital certificates to provide authentication, integrity, confidentiality, and non-repudiation. It extends the MIME standard by adding security features.

MIME

MIME (Multipurpose Internet Mail Extensions) is an extension of the traditional e-mail format that allows transmission of:

- Images
- Audio files
- Video files
- Documents
- Multimedia content

S/MIME adds security to MIME messages.

Features of S/MIME

1. **Confidentiality:** Ensures that only the intended recipient can read the message through encryption.
2. **Authentication:** Verifies the sender's identity using digital certificates.
3. **Integrity:** Detects any unauthorized modification of the message.
4. **Non-Repudiation:** Provides proof that the sender actually sent the message.
5. **Certificate-Based Trust:** Uses Certificate Authorities (CA) to establish trust between users.

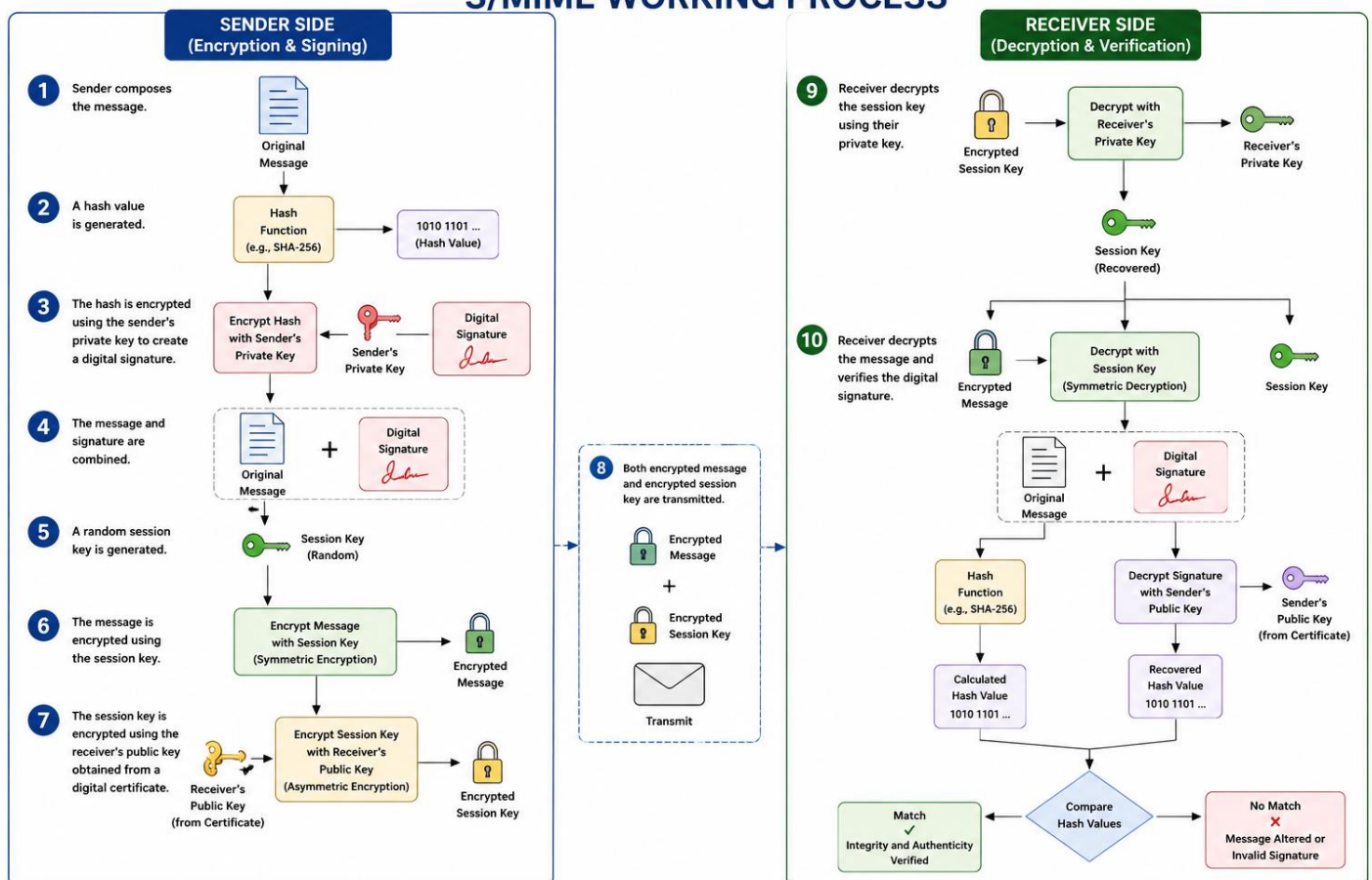
S/MIME Services

1. **Enveloped Data:** Provides confidentiality by encrypting the entire message.
2. **Signed Data:** Provides authentication and integrity through digital signatures.
3. **Clear Signed Data:** The message remains readable while the signature is attached separately.
4. **Signed and Enveloped Data:** Provides both encryption and digital signature protection.

S/MIME Working Process

1. Sender composes the message.
2. A hash value is generated.
3. The hash is encrypted using the sender's private key to create a digital signature.
4. The message and signature are combined.

S/MIME WORKING PROCESS



5. A random session key is generated.
6. The message is encrypted using the session key.
7. The session key is encrypted using the receiver's public key obtained from a digital certificate.
8. Both encrypted message and encrypted session key are transmitted.

9. Receiver decrypts the session key using their private key.
10. Receiver decrypts the message and verifies the digital signature.

Advantages of S/MIME

- Industry-standard e-mail security solution.
- Easy integration with modern e-mail clients.
- Uses trusted Certificate Authorities.
- Provides confidentiality, authentication, and integrity.
- Suitable for enterprise environments.

Disadvantages of S/MIME

- Requires digital certificates.
- Certificate management can be expensive.
- Depends on Certificate Authorities.

Feature	PGP (Pretty Good Privacy)	S/MIME (Secure/Multipurpose Internet Mail Extensions)
Standard Type	Open standard	IETF standard
Key Management	User-managed web of trust	Certificate Authority (CA)-based trust model
Digital Certificates	Optional	Mandatory
Trust Model	Web of Trust	Hierarchical PKI (Public Key Infrastructure)
Authentication	Digital Signature	Digital Signature
Confidentiality	Message Encryption	Message Encryption
Integrity	Hash Functions and Signatures	Hash Functions and Signatures
Encryption Algorithm	RSA, IDEA, CAST, AES, etc.	RSA, AES, 3DES, etc.
Signature Algorithm	RSA, DSA	RSA, DSA, ECDSA
Message Format	Uses PGP message format	Uses MIME format
Certificate Storage	Local key ring	X.509 Certificates
Ease of Deployment	More complex for organizations	Easier in enterprise environments
Email Client Support	Requires PGP-compatible software	Built into many email clients
Scalability	Better for individuals and small groups	Better for large organizations
Cost	Usually free/open source	May require CA-issued certificates
Security Management	User controls trust relationships	Centralized certificate management
Main Usage	Personal secure email communication	Enterprise and business email security
Examples	GnuPG, OpenPGP	Microsoft Outlook S/MIME, Apple Mail S/MIME